

Freeway is a game where you control a chicken with actions of UP, DOWN, or NO-OP (“no-operation” or do nothing). The objective is to move the chicken across the freeway, avoiding oncoming traffic, to get to the other side, where you get a reward of +1. If you don’t get all three chickens across the road in a limited amount of time, you lose the game and get a negative reward.

In most cases in this book, we train our DRL agents using the raw pixel representation of the game and thus use convolutional layers in our neural network. In this case, though, we’re introducing new complexity by making a distributional DQN, so we’ll avoid convolutional layers to keep the focus on the topic at hand and keep the training efficient.

The RAM version of each game is essentially a compressed representation of the game in the form of a 128-element vector (the positions and velocities of each game character, etc.). A 128-element vector is small enough to process through a few fully connected (dense) layers. Once you